

Lab 3: Lifecycle Selection & Increment Plan

Formative (ungraded) · Moodle-enabled · output feeds Task 2 Part 2 directly

Item	Details
Linked to	Week 3 lecture — Software Process Models (ULO2, ULO3)
Work mode	Individually, on the SAME client scenario you chose for Task 2
Time	About 90 minutes
Submission	Completed worksheet uploaded to Moodle AND committed to week03/ in your Git portfolio
Why it matters	Section C of this worksheet is a first draft of Task 2 section 2.3 (work schedule). Do it properly now and Week 8 gets much easier.

Purpose

In the lecture you met six ways to organise the same four activities, and six questions that decide which one fits. Now you apply that decision to a real client — yours. There is no single correct model; the marks in Task 2 (and the exams) come from the quality of your justification.

Part A — Apply the six selection questions (30 min)

For each question, answer for YOUR client scenario (Kina Ridge Primary School or Waghi Valley Produce Cooperative) and say what it pushes you toward.

#	Selection question	Your answer for this client	Pushes toward...
1	How well understood are the requirements?	The school knows they need basics like attendance and report cards, but they haven't fully thought through the details of parent notifications or how term grading should work until they see a working example.	Iterative / Agile (We need to build a little, show them, and let them tell us what to change).
2	Can the client engage continuously?	Not really. Teachers are swamped during the day and can only meet briefly after 3 PM. They are also completely unavailable during exam weeks or school holidays.	Waterfall (Less need for daily chat) OR Incremental (We only need them at specific handover times).

3	What does the contract or funding demand?	The project is funded by a community grant that releases money only when specific milestones are hit, like "Enrollment Module Done." They need clear proof of completion	Waterfall / Phased (Clear, documented milestones make getting paid easier).
4	How big is the team, and where is it?	It's just us (3 students), and we are all on campus together. We can talk face-to-face anytime and change plans quickly without needing heavy paperwork.	Agile / Scrum (Small, close teams work best this way).
5	What is the cost of failure?	Very high. If we calculate grades wrong or lose attendance records, parents will lose trust in the school, and the school could get in trouble with education authorities.	V-Model / Waterfall (We need strict testing to ensure zero errors).
6	How fast must value appear for this client?	Pretty fast. They need the Enrollment system running now for the next term intake in 6 weeks, but they can wait a bit longer for the complex grading features.	Incremental (Give them the urgent stuff first, add the rest later).

Part B — Choose and justify (15 min)

Chosen lifecycle: Incremental Model (blended with Waterfall-style milestones) (may be a blend — say so explicitly)

In 150–200 words, justify the choice using your Part A answers. Name at least one factor that argues AGAINST your choice and say how you will manage it.

Justification: I chose the Incremental Model because it fits the school's immediate needs best. The biggest reason is urgency: the school desperately needs the Enrollment module ready for the upcoming term, and this model lets us deliver that specific piece quickly without waiting for the whole system to be finished. Also, since the teachers aren't totally sure what they want yet (Question 1), building in chunks allows them to try the Enrollment system first and give us feedback before we build the harder parts like Grading.

The main challenge against this choice is the funding rule (Question 3). The grant committee wants rigid, "finished" milestones to release money, which sounds more like Waterfall. To handle this, I'll use a hybrid approach: my team will work flexibly and iteratively to get the software right, but we will package our work into formal "Phases" for the committee. This way, we satisfy the strict reporting rules for funding while still keeping the flexibility to adapt the software based on what the teachers actually need. We will also be extra careful with testing on each increment to avoid the high cost of failure (Question 5).

Blends are professional, not a cop-out. "Incremental delivery internally, phase milestones reported to the board" is a legitimate and common answer in PNG work — provided you explain how the two fit together.

Part C — Plan three increments (35 min)

Split your client's system into three increments. Increment 1 must be genuinely usable on its own — something the client could put to work the day you hand it over.

Increment	What it delivers	Why THIS order (risk or value?)	Rough duration & why
1	Student Enrollment & Basic Profile (Register new students, assign them to classes, print ID lists).	Highest Value. The school literally cannot start the new term properly without knowing who is enrolled. This is the foundation everything else sits on.	5 Weeks. We have to finish before the term starts. It's also the simplest part, so it's a safe place to start.
2	Daily Attendance & Term Reports (Mark the roll each morning, view absence stats, print simple reports).	High Risk / Legal Need. Tracking attendance is a legal requirement. Doing this second lets us check if the student data from Increment 1 is working correctly before adding more complexity.	6 Weeks. We need to spend time making the interface super fast so teachers can mark the roll quickly in the morning rush.
3	Gradebook & Advanced Analytics (Enter marks, auto-calculate GPA, generate official report cards).	Complexity. This involves tricky math and logic. By doing it last, we can make sure the database is stable based on real use from the first two parts.	7 Weeks. This needs the most testing to ensure no calculation errors happen on report cards.

C4. Delivery-condition check

For each increment, note one PNG delivery condition that affects when or how you can deliver it (client travel, power, connectivity, harvest or school-term timing, committee approval cycles).

Condition notes	
Increment	Condition notes (PNG delivery context)
1	School Term Timing: This must be delivered before Week 1 of Term 3. If we are late, the school will just go back to using paper books, and they might never switch to our system.
2	Power & Connectivity: The highlands often have power cuts. The system must work offline on the teacher's laptop and sync to the principal's computer later when power is back.
3	Committee Meetings: The Board of Management only meets once a month to approve report card formats. We have to time our delivery to match their meeting schedule, or we'll wait weeks for approval.

Part D — Reflect (10 min)

- If the client suddenly demanded ALL three increments by the same original deadline, which corner of the dependency triangle would you push back on, and what exactly would you say to them?
- Which of your three increments is most likely to change after the client sees increment 1? What does that tell you about how firmly to specify it now?

Your answers

Q1: I would push back on Scope (the features). What I would say: "We can hit that deadline, but not with everything. If we rush the Gradebook (Increment 3) to be ready at the same time as Enrollment, we risk making calculation mistakes that could hurt the school's reputation. I propose we deliver the Enrollment and Attendance modules fully tested by the deadline, and give you a simplified, manual version of the Gradebook first. We can then finish the full automated Gradebook just two weeks later. This keeps your data safe and ensures the critical parts work perfectly on day one."

Q2: Increment 3 (Gradebook) is the most likely to change. Once teachers start using the Enrollment and Attendance tools, they will realize new things they need. For example, they might say, "Oh, we need to separate 'excused' absences from 'unexcused' ones before calculating grades," which changes how the Gradebook needs to work. What this tells me: I should not lock down the detailed rules or screen designs for the Gradebook right now. I will just specify the basic data needs (like "must store a score") and leave the rest flexible so we can adjust it easily after seeing how they use the first two increments.

Submitting

1. Save as Lab3_<YourName>.docx and upload to Moodle → Week 3 → Lab 3.
2. Copy it into week03/ in your portfolio and commit:

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git add week03/ && git commit -m "Add Lab 3 lifecycle selection and increment plan"
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3. Keep it open when you start Task 2 Part 2 — Parts B and C become sections 2.3 and 2.4.